UNIT – V

OBJECTIVE:

To familiarize the students with the topics of combustion , chemical reactions of fuels combustion analysis.

COMBUSTION:

Combustion is the chemical combination of the combustible constituents of a fuel with oxygen. During this chemical reaction carbon, hydrogen and sulphur present in the fuel combine vapidly with oxygen to form carbon dioxide (or) carbon monoxide (co), sulphur dioxide (so_2) and water (H_2o) with the release of large amount of energy. To begin this process, temperature of the fuel must be raised to a certain value known as ignition point of the fuel. The fuel and oxidizer which supplies the necessary oxygen are known as reactants and the constituents resulting from the combustion process are called the proudest.

STOICHIOMETRY:

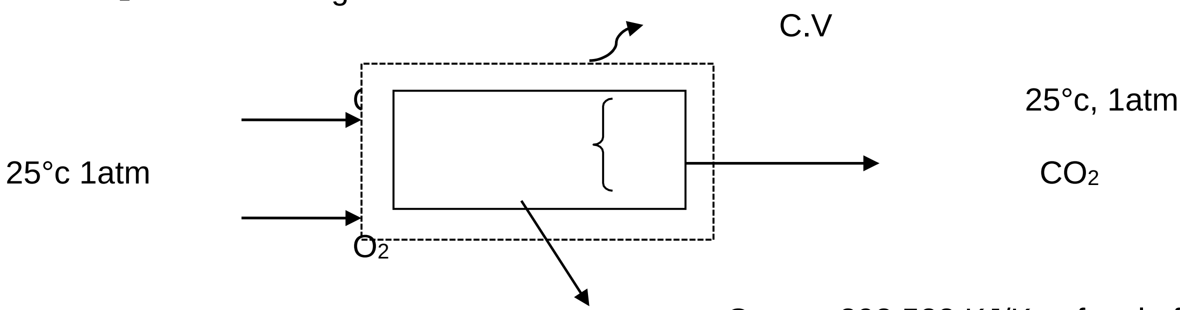
The minimum amount of air which provides sufficient oxygen for the complete combustion of all the elements like carbon, hydrogen etc. which may oxidize is called the theoretical or stoichiometric air. There is no oxygen in the products when complete combustion oxidation is a achieved with this theoretical air.

Theoretical or minimum air required for complete combustion of 1 kg of fuel

$$= \frac{100}{23}\left[\left(\frac{8}{3}c + 8H_2 + S\right) - o\right] kg.$$

ENTHALPY OF FORMATION:

Let us consider the steady state steady flow combustion of carbon and oxygen to form Co_2 as shown in figure.



$Q_{C.V}$ = -393,522 KJ/Kg of mol of CO_2

- Let the carbon and oxygen enter the control volume at 25°c and 1 atm, pressure and the heat transfer be such that the product CO_2 leaves at 25°c, 1atm pressure
- The measure value of heat transfer is -393,522 KJ per kg of CO_2 formed
- If H_R and H_P refer to the total enthalpy of the reactants and products respectively.

Then the first law applied to the reaction

$C + o_2 \longrightarrow Co_2$ gives

$$\boxed{H_R + Q_{c.v} = H_p}$$

Generally all the reactants and products in a reaction , the equation may be written as

$$\boxed{\sum_R n_i \bar{h}_i + Q_{c.v} = \sum_p n_e \bar{h}_e}$$

Where R and P refer to the reactants and products respectively.

- The enthalpy of all the elements at the standard reference state of 25°c ,1 atm is assigned the value of zero
- In carbon- oxygen reaction, $H_R = 0$

i.e. $H_P = Q_{c.v} = -393522$ KJ/Kgmol

It is also denoted by $\bar{h}_f^{\circ}$

⇨ $\bar{h}_f^{\circ}$ = -393522 KJ/Kgmol

But in most cases, the reactants and products are not at 25°c, 1 atm. Therefore, the change in enthalpy between 25°c, 1 atm and the given state must be known. Thus the enthalpy at any temperature and pressure, $\bar{h}_{T,p}$ is

$$\boxed{\bar{h}_{T,p} = \bar{h}_f^{\circ} + \Delta\bar{h}}$$

Where $\Delta\bar{h}$ referents the difference in enthalpy between any given state and the enthalpy at 298.15K, 1atm.

ADIABATIC FLAME TEMPERATURE:

Consider a given combustion process that takes place adiabatically and with no work or changes in K.E or potential energy involved. The temperature of the products is referred to as the adiabatic Flame temperature.

With the assumption of no work and no changes in kinetic or potential energy, this is the maximum temperature that can be achieved for the given reactants because any heat transfer from the reacting substances and any incomplete combustion would tend to lower the temperature of the products.

For a given fuel and given pressure and temp of reactants, the maximum adiabatic flame temperature that can be achieved is with a stoichiometric mixture. The Adiabatic flame temperature can be controlled by the amount of excess air that is used.

The energy equation is

$$H_R = H_p$$

$$\sum_R n_i \bar{h}_i = \sum_p n_e \bar{h}_e$$

Or

$$\sum_R n_i [\bar{h}^0_f + \Delta\bar{h}]_i = \sum_p n_e [\bar{h}^0_f + \Delta\bar{h}]_e$$

For example:

In gas turbine, where the maximum permissible temperature is determined by metallurgical consideration in the turbine is determine by metallurgical consideration in the turbine and close control of the temp. Of the products is essential.

ENTHALPY OF REACTION AND HEATING VALUE:

The enthalpy of combustion is defined as the difference between the enthalpy of the products and the enthalpy of reactants when complete combustion occurs at a given temperature and pressure

$$\bar{h}_{RP} = H_p - H_R$$

Or

$$\bar{h}_{RP} = \sum_R n_e [\bar{h} + \Delta\bar{h}]_e - \sum_p n_i [\bar{h} + \Delta\bar{h}]_i$$

Where $\bar{h}_{RP}$ is the enthalpy of combustion (KJ/Kg or KJ/mol) of the fuel

The internal energy of combustion, U_{RP} is defined in

$$\bar{u}_{RP} = U_p - U_R$$

$$= \sum_R n_e [\bar{h} + \Delta\bar{h}]_e - \sum_p n_i [\bar{h} + \Delta\bar{h}]_i$$

If all the gaseous constituent are considered ideal gases and the volume of liquid and solid considered is assume to be negligible compared to gaseous volume.

$$\bar{U}_{RP} = \bar{h}_{RP} - \bar{R}T\ (\eta_{\text{gaseous products}} - \eta_{\text{gaseous Reactants}})$$

- In the case of a constant pressure or steady flow process, the negative of the enthalpy of combustion is frequently called the heating value at constant pressure, which represents the heat transferred from the chamber during combustion at const. pressure.
- Similarly, the negative of the internal energy of combustion is called as the heating value at constant volume in the case of combustion, because it represents the amount of heat transfer in the const volume process.
- The higher heating value (HHV) or higher calorific value (HCV) is the heat transferred when H_2O in the products is in the liquid phase.
- The lower heating value (LHV) or lower calorific value (LCV) is the heat transferred when H_2O in the products is in the vapour phase.

$$\boxed{LHV = HHV - m\ H_2O.\ hfg}$$

Where mH_2O is the mass of water formed in the reaction.

AIR – FUEL RATIO

Air – fuel ratio is the main quantity in the analysis of combustion process. It is defined as the ratio of the mass of air to the mass of fuel during a combustion process. That is

$$\text{Air – fuel Ratio} = \frac{Mair}{Mfuel}$$

EFFECTS ON SUPPLY OF EXCESS AIR:

If the air supply is less than the stoichiometric amount, some of the carbon will become carbon monoxide or will remain as carbon particle in the product gas.

If the air supply is greater than the stoichimometric amount, the product of combustion will contain free oxygen.

COMBUSTION REACTION:

During combustion, the fuel is oxidized so that the carbon becomes co_2 or co, Hydrogen becomes H_2o and sculpture becomes so_2.

Consider 1 kmol of carbon combining with 1 kmd of oxygen to form 1 kmol of carbon dioxide.

$C + o_2 \Rightarrow co2$

In terms of main

1 kmol of c => 12 kg, 1 kmol of $_{02}$ => 32 kg

1kmol of co_2 => 44kg

Consider carbon combining with oxygen to for carbon monoxide.

$2c + o2 \Rightarrow 2co$

2kmol of carbon combines with 1 kmol of oxygen to form 2 kmol of carbon monoxide.

2 kmol of c => 24 kg, 1 kmol of o_2 => 32 kg

2 kmol of co => 56 kg.

Oxidation of sulphur gives.

$S + o_2 \Rightarrow so_2$

1 kmol of sulphur combines with 1 kmol of oxygen to form of sculpture dioxide.

1 kmol of s => 32 kg, 1 km of o_2 => 32 kg.

1 km01 of so_2 => 64 kg.

ENTHALPY OF COMBUSTION:

The internal energy of combustion is defined as the difference between the internal energy of the product and the internal energy of reactants when complete combustion occurs at a given temperation and pressure.

$$u_{RP} = U_p - U_R$$

$$= \sum_P ne\left[h \circ f + \Delta h - p\nu\right]e - \sum_R ni\left[hf \circ + \Delta h - p\nu\right]i$$

HEATING VALVE OF FUELS:

The heating value of a fuel is the amount of heat transferred from the fuel when it is burned in stochiometric proportions of air in a steady – flow process such that the reactants are at the standard reference state and the products are returned to the standard reference state.

The heating value is numerically equal to the enthalpy of combustion, but it has the opposite sign.

There two types of calorific values namely:-

a) Higher calorific value
b) Lower calorific value

HIGHER CALORIFIC VALUE: (HCV)

Higher calorific value (HCV) is the quantity of heat obtained by the complete combustion of unit mass of the fuel, when the products of combustion are cooled down to the temperature of the surroundings. In this case any water vapour formed by combustion is condensed and the heat is recovered from the products of combustion.

LOWER CALORIFIC VALUE: (LCV)

Lower calorific value is the quantity of heat abstained by the complete combustion of unit mass of the fuel, when the products of combustion are not sufficiently cooled down to condense the steam formed during combustion.

LUV = HCV – Latent heat of steam formed
HCV – 2442 x steam formed (kj/kg)

ADIABATIC FLAME TEMPERATURE:

The maximum temperature that products can reach during a combustion process which is adiabatic involving no work and negligible changes is kinetic and potential energy. Mathematically the energy balance gives

$$\phi = W + Hp - HR$$

$$But \phi - W = 0$$

$$==> HP = HR$$

$$(or)$$

$$\sum_R nihi = \sum_P nehe$$

$$==> \sum_R ni\left[hf\circ + \Delta h\right]i = \sum_P ne\left[hf\circ + \Delta h\right]e$$

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CHEMICAL EXERGY:

The chemical exergy is thus defined as the maximum theoretical work that could be developed by the combine system. Thus for a given system at a specific state.

Total exergy = thermo chemical exergy + chemical exergy.

SECOND LAW EFFICIENCY OF A REACTIVE SYSTEM.

For a fuel at To, Po, the chemical exergy is the maximum theoretical work that could be obtained through reaction with environmental substances. However due to various irreversibility like friction and heat loss, the actual work obtained is only a fraction of this maximum theoretical work.

The second law efficiency may thus be defined as the ratio of.

$$\eta II = \frac{ActualWorkdone}{Maximumtheoreticalwork}$$

$$= \frac{Wcv}{mfuel \times ach}$$

PROBLEMS:

1. The products of combustion of an unknown hydrocarbon $C_x H_y$ have the following composition as measured by an orsat apparatus:

   CO_2 8.0 % , CO 0.9% , O_2 8.8 % , and N_2 82.3%

Determine : a) the composition of the fuel, b) the air fuel ratio, C) the % of excess air used

SOLUTION:

Let a moles of oxygen be supplied per mole of fuel. The chemical reaction can be written as follows.

$C_x H_y + aO_2 + 3.76\ a\ N_2$

$= \quad 8\ CO_2$, $0.9\ CO$, $8.8\ O_2$, and $82.3\ N_2 + y / 2\ H_2O$

Oxygen balance gives

$2a = 16 + 0.9 + 17.6 + y/2$ ………… 1

Nitrogen balance gives

$3.76a = 82.3$

$a = 21.89$

by substituting the value of a in eqn 1

$y = 18.5$

carbon balance gives $x = 8 + 0.9 = 8.9$

therefore, the chemical formula of the fuel is $C_{8.9} H_{18.5}$ The composition of the fuel is

$$\%\ \text{carbon} = \frac{8.9 \text{ X } 12}{8.9 \text{ X } 12 + 18.5 \text{ X} 1} \text{X } 100 = 85.23\ \%$$

% of Hydrogen = 14.77 %

B) $\text{air fuel ratio} = \frac{32a+3.76a \text{ X } 28}{12x+y}$

$$= \frac{21.89\ (32+3.76\ X\ 28)}{12X\ 8.9+18.5} = \frac{3005}{125.3} = 24$$

C) Excess air used:

$$\frac{8.8 \text{ x } 32}{21.89 \text{ x } 32 - 8.8 \text{ x } 32} \text{ x } 100 = 67.22\ \%$$

2. Determine the heat transfer per kg mol of fuel for the following reaction

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$$

The reactants and products are each at a total pressure of 100Kpa and 25°C

SOLUTION:

By the first law

$$Q_{C.V} + \sum_P n\,\hbar_i = \sum_P n\,\hbar_e$$

From table C in the appendix

$$\sum_R n_e \hbar_e = (\hbar_F)_{CO2} + 2(\hbar)_f O_2$$

= - 393, 522 + 2(- 285,838) = - 965,198 KJ

$Q_{c.v}$= -890,325 KJ